

Aim:

You are a bioinformatics scientist at the Indian Institute of Science of Technology (IIST), Bengaluru, working in a cutting-edge genetic research laboratory. Scientists are comparing DNA samples from two species found in the Western Ghats to identify common evolutionary patterns.

Your task is to determine the length of the Longest Common Subsequence (LCS) between two DNA sequences, which reveals shared genetic traits and evolutionary links.

Input Format:

- The first line contains a string X , representing the DNA sequence of the first species.
- The second line contains a string Y , representing the DNA sequence of the second species.

Output Format:

- Print a single integer representing the length of the Longest Common Subsequence (LCS) between X and Y .
- If no common subsequence exists, print 0.

Note:

- The output must end with a newline character ($\backslash n$) to ensure proper formatting.
- The comparison is case-sensitive; uppercase and lowercase characters are treated as different.
- Partial code is provided. Complete the **max()** function and LCS logic.

Source Code:

sequenceAnalysis.c

```
#include <stdio.h>
#include <string.h>

int max(int a, int b) {
    return(a>b)?a:b;
    // Complete this function
}

int main() {
    char X[1005], Y[1005];
    scanf("%s", X);
    scanf("%s", Y);
    int m = strlen(X);
    int n = strlen(Y);
    int L[m + 1][n + 1];

    // Complete the LCS logic here
    for(int i=0;i<=m;i++){
        for(int j=0;j<=n;j++){
            if(i==0 || j==0)
                L[i][j]=0;
            else if (X[i-1]==Y[j-1])
```

```
        L[i][j]=max(L[i-1][j],L[i][j-1]);
    }
}
printf("%d\n",L[m][n]);
return 0;
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
ABCB DAB
BDCAB
4

Test Case - 2
User Output
ABC
DEF
0